Segmentation, Representation Geometry, and Analytic Aggregate Risk in Motor Insurance Pricing

A computational exploration on the freMTPL2 benchmark

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Abstract

This note explores the interaction between segmentation, categorical representation, and analytic aggregate risk modelling in motor insurance pricing. Starting from the freMTPL2 benchmark dataset, several segmentation approaches are combined with Tweedie GLM pricing and analytic compound Poisson aggregation.

Two questions motivated the work. The first is operational: whether analytic aggregate methods based on convolution can provide a practical alternative to large-scale Monte Carlo simulation for portfolio risk evaluation. The second is representational: whether the Burt matrix provides a useful latent geometry for categorical insurance variables compared with classical one-hot encoding.

The experiments suggest that soft segmentation through Gaussian mixture models improves pricing discrimination relative to a plain GLM baseline, while Burt-based representations appear to produce less extreme tail behaviour and improved top-decile calibration. The work is exploratory rather than definitive, but it points toward an interesting interaction between representation geometry, computational tractability, and aggregate actuarial behaviour.

1. Motivation

This work grew out of practical experimentation rather than actuarial research in the formal academic sense. My background is primarily mathematical and computational, with a long-standing interest in legacy modelling systems, numerical workflows, and implementation structure.

The initial question was operational. In experimenting with aggregate motor insurance modelling in Python, I found that classical Monte Carlo simulation became surprisingly fragile once portfolios were segmented into heterogeneous groups with differing severity behaviour. Runtime increased quickly, while tail estimates exhibited visible simulation noise unless the number of simulations became very large.

This naturally led back toward analytic collective risk methods. The Python package aggregate makes compound Poisson convolution accessible through FFT methods, allowing exact or near-exact aggregate distributions to be computed rapidly and reproducibly.

A second thread came from reading a published paper by Hainaut, Trufin and Denuit on insurance risk classification using Generalized Gaussian Process Regression (GGPR), available on UCLouvain's BOREAL repository, which introduced the Burt matrix as a continuous embedding of categorical rating factors. The object itself is mathematically elegant. If Z denotes the complete indicator matrix of categorical modalities, the Burt matrix is defined as $B = Z^T Z$.

The resulting geometry encodes co-occurrence structure between modalities and induces a latent continuous representation of categorical profiles. The question was not only whether this representation could improve prediction, but whether it changed the behaviour of segmentation and aggregate portfolio structure once models were implemented end-to-end.

2. Compound Poisson Aggregate Structure

2.1 Segment-level aggregate

For segment j, claim frequency is represented by a Poisson distribution with intensity λ_j. Severities X_{ij} follow a segment-specific severity law F_j. In the experiments described here, F_j is modelled as lognormal and parameterised by mean μ_j and coefficient of variation CV_j.

$$S_j = \sum_{i=1}^{N_j} X_{ij}$$

The random variable S_j is a compound Poisson aggregate, whose distribution is obtained by FFT convolution of the discretised severity law.

The implementation uses the *aggregate* package and its *DecL* specification language. Conceptually, the segment parameters enter as vectors: one intensity vector for claim counts, and corresponding vectors for severity means and severity coefficients of variation.

agg Portfolio [λ_1 … λ_k] claims · sev lognorm [μ_1 … μ_k] cv [CV_1 … CV_k] poisson

2.2 Portfolio aggregate

When the portfolio is segmented into k independent groups, the total aggregate is the sum of the segment aggregates. This is a convolution of independent compound Poisson aggregates, not a statistical mixture of aggregate distributions.

$$S = S_1 + S_2 + \cdots + S_k$$

Under independence, the total remains analytically tractable. Equivalently, it may be viewed as a compound Poisson aggregate with total intensity equal to the sum of the segment intensities and a severity distribution given by the intensity-weighted mixture of the segment severity laws.

The aggregate distribution is analytically evaluated through FFT convolution rather than Monte Carlo simulation. In practice, this allows VaR and TVaR figures to be computed rapidly and without simulation noise.

3. Segmentation and Representation

3.1 Experimental design

The experiments combine Tweedie GLM pricing, Gaussian mixture segmentation, K-means segmentation, and analytic aggregate modelling. The objective is not prediction alone. Three distinct questions are explored simultaneously:

- Does segmentation improve pricing discrimination?
- Does representation geometry affect calibration behaviour?
- Does segmentation structure propagate into aggregate tail behaviour?

Two broad feature geometries are evaluated. In the classical representation, categorical rating variables are introduced directly through one-hot encoding. In the Burt representation, categorical variables are embedded into a latent continuous geometry derived from the Burt matrix.

3.2 Choice of segmentation dimension

For comparability across experiments, a fixed $k = 5$ segmentation was retained throughout most analyses. $k=5$ was selected from elbow diagnostics on K-means inertia curves and retained as a fixed reference across representation families.

For enriched Gaussian mixture experiments, BIC-selected values of k were also evaluated independently within each feature geometry. This keeps the fixed-k comparisons interpretable while still allowing the GMM benchmark to reflect a data-driven segmentation dimension.

3.3 The Burt representation

The clustering geometry is derived from the Burt representation introduced earlier. The off-diagonal blocks count co-occurrences between modalities. The construction is closely related to Multiple Correspondence Analysis (MCA) and correspondence geometry. Rather than treating categorical modalities as independent binary indicators, the Burt structure embeds them into a continuous latent space reflecting their empirical co-occurrence frequencies. Modalities that frequently appear together in the portfolio are positioned closer in the induced geometry.

From a segmentation perspective, this produces a smoother relational representation than sparse one-hot encoding. The resulting latent geometry may reduce abrupt partition boundaries and produce more gradual transitions between neighbouring policyholder profiles. After normalisation and eigendecomposition, policies can be embedded into a continuous latent space where nearby points correspond to similar categorical profiles.

Two Burt variants are explored. In the Burt-replacing design, the original categorical factors are removed from the pricing model and replaced by Burt coordinates. In the Burt-augmented design, the original factors are retained alongside the Burt coordinates. In the present experiments, the augmented design provides almost no additional lift over the replacing design, suggesting that the Burt coordinates capture most of the predictive structure carried by the original categorical variables in this setting.

4. Results

4.1 Pricing behaviour

The strongest discrimination lift comes from GMM segmentation applied to the *enriched* geometry combining categorical structure with predicted frequency and severity. However, the Burt-based representations exhibit noticeably better calibration behaviour in the upper deciles. This trade-off ultimately appeared more informative than the raw Gini improvements themselves.

Experiment	Series	Gini	Deviance/exp.	Δ vs baseline	Top-decile ratio
A5 — Classical GMM enriched k=7	A	0.4241	61.276	+0.0552	1.139
A3 — Classical GMM enriched k=5	A	0.4227	61.290	+0.0539	1.090
B5 — Burt GMM enriched k=8	B	0.3796	61.336	+0.0107	1.032
C3 — Burt-augmented GMM enriched k=5	C	0.3815	61.290	+0.0127	1.032
A0 — Classical plain GLM baseline	A	0.3688	61.691	0.0000	1.032

The classical enriched GMM models produce higher separation between risks, but also appear to widen the aggregate tail. The Burt-based models are less aggressive from a pricing perspective and are associated with more moderate aggregate tail metrics and improved top-decile calibration in these experiments.

4.2 Aggregate behaviour

Aggregate distributions were evaluated analytically through FFT convolution. The Burt-based segmentation produced slightly lower aggregate coefficient of variation, lower 99% VaR, lower 99% TVaR, and improved top-decile calibration. The numerical differences are not dramatic in absolute terms, but they appear directionally consistent.

Metric	A5 — Classical	B5 — Burt	Difference
Gini coefficient	0.4241	0.3796	+0.0445 (A5)
Tweedie deviance / exp.	**61.276**	61.336	+0.060 (A5)
Top-decile ratio	1.139	1.032	−0.107 (B5)
Aggregate CV	0.01053	0.01000	−0.00053
VaR 99%	61,395,800	61,322,700	−73,100
TVaR 99%	61,618,566	61,533,968	−84,598
Aggregate validation	not unreasonable	not unreasonable	—

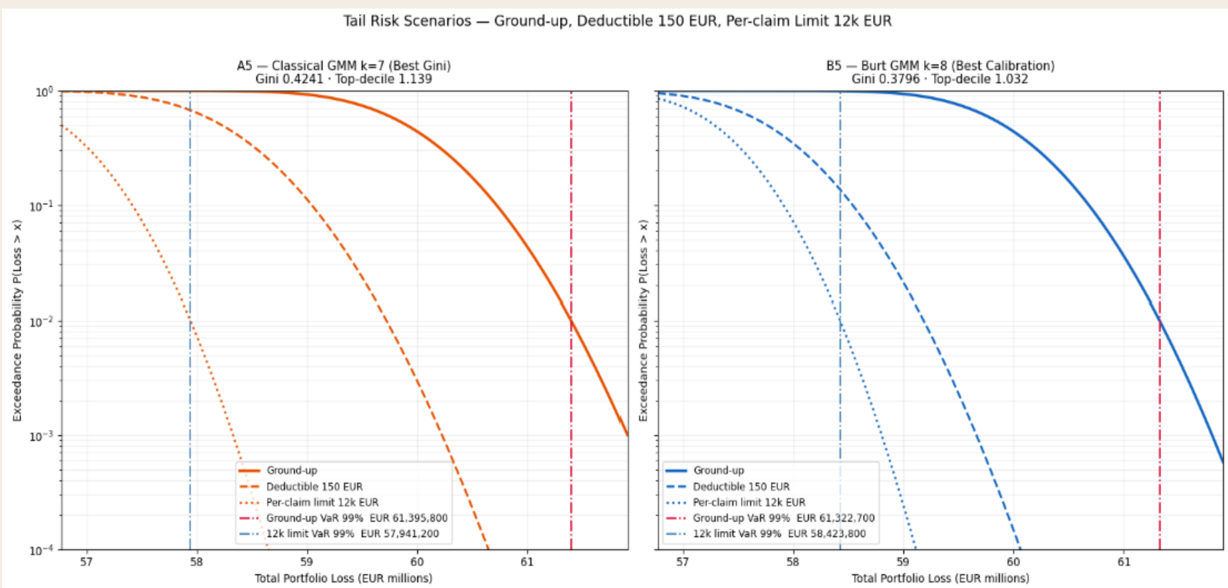

Figure 1 —Each segment contributes an independent compound Poisson sub-aggregate; the total portfolio loss is their convolution, evaluated analytically with FFT.

Exceedance probability curves for ground-up, deductible, and per-claim-limit scenarios. Vertical markers indicate the 95% and 99% VaR of the ground-up aggregate distribution only.

The figure illustrates how segmentation structure propagates into aggregate tail behaviour under different coverage layers.

4.3 Coverage layer analysis

The coverage layer analysis reveals a reinsurance-dependent reversal. Under a deductible of 150 EUR, B5 produces a lower aggregate mean (€470,871 lower) and lower TVaR 99% (€549,654 lower) than A5 — the better tail calibration of the Burt representation translates directly into a capital advantage. Under a per-claim limit of 12,000 EUR, the ranking reverses: B5 produces a higher aggregate mean (€509,634 higher) and higher TVaR 99% (€478,305 higher). The per-claim limit **caps** the large individual claims that drive B5's heavier aggregate tail, while A5's sharper risk segmentation delivers better aggregate behaviour once the severity upside is truncated. The choice between the two representations is therefore not purely a *discrimination* versus *calibration* trade-off — it depends on the reinsurance structure in place.

5. Interpretation

The experiments suggest a tension between discrimination and moderation of aggregate tail behaviour. More aggressive segmentation improves predictive separation but may also amplify tail concentration and calibration distortion. Smoother latent representations appear to moderate some of this behaviour.

From an actuarial perspective, this raises an operational question: should pricing models be optimised purely for discrimination metrics such as Gini, or should aggregate portfolio behaviour also be treated as part of the optimisation target?

This becomes particularly relevant once aggregate capital measures such as VaR or TVaR are integrated directly into pricing workflows.

More broadly, the experiments suggest that actuarial modelling is not only a problem of statistical estimation. Representation structure, computational tractability, segmentation geometry, and aggregate behaviour interact quite strongly once models are implemented operationally.

An important outcome of the experiments is that actuarial predictive enrichment appears to dominate purely relational categorical geometry for pricing discrimination. The enriched classical representation — combining one-hot categorical structure with predicted frequency and predicted severity — consistently produces the strongest Gini performance. This suggests that supervised actuarial latent structure carries more pricing information in this benchmark than unsupervised relational embedding alone. This does not imply that relational geometry is uninformative; rather, it suggests that the incremental pricing signal carried by the Burt embedding overlaps substantially with the signal already captured by the supervised actuarial enrichment process.

6. Limitations

The experiments remain exploratory and several simplifying assumptions are used.

- Segment severities are approximated through *lognormal* distributions compatible with the analytic convolution framework used in the Python aggregate workflow.
- Segment-level severity coefficients of variation are estimated empirically from observed claim amounts using weighted lognormal MLE, with soft GMM membership weights restricted to claimants. The empirical CVs range from approximately 1.11 to 1.83 across segments and have a material impact on aggregate tail figures.
- Segment independence is assumed.
- Tail dependence is not modelled.
- The work focuses on benchmark data rather than production insurance portfolios.
- The pricing framework remains GLM-based; non-linear learners such as gradient boosting may behave differently. This extension has since been carried out and is reported separately in the *Gradient Boosting Extension* (June 2026).
- The PCA stabilisation applied to the Burt geometry is not interpreted as part of the representation itself, but as a numerical regularisation step required for full-covariance Gaussian mixture estimation in high-dimensional latent spaces.

These limitations mean the results should be interpreted primarily as structural observations rather than definitive actuarial conclusions.

7. Conclusions

Seven conclusions seem worth retaining.

- Soft segmentation through Gaussian mixture models improves pricing discrimination by approximately 5.5 Gini points over the plain GLM baseline. Hard K-means clustering on the same enriched geometry produces no improvement and in one configuration falls below the baseline — confirming that soft probabilistic assignment is essential to exploit the enriched feature space.
- Tweedie deviance per exposure is added as a second fit-quality metric, following the recommendation by Hainaut & Denuit (Detra Note 2025-2) that deviance is the standard actuarial goodness-of-fit criterion for models in the exponential dispersion family. A5 achieves the lowest deviance (61.276 per exposure unit) across all 21 experiments — the same experiment that wins on Gini. Deviance and Gini are aligned within the A-series. However, across representation families the overall correlation is weak and not statistically significant: the B and C-series cluster at competitive deviance values despite substantially lower Gini. The top-decile calibration ratio is a genuinely orthogonal dimension that neither Gini nor deviance captures.
- The best-calibrated Burt experiment (B5, GMM enriched k=8, BIC-confirmed) achieves a top-decile ratio of 1.032 against A5's 1.139 — a 10.7 percentage point improvement — and produces a TVaR 99% approximately €85,000 lower. Both A5 and B5 pass the aggregate validator cleanly. The Gini gap of 4.5 points is a property of the Tweedie GLM, which is architecturally optimised for categorical contrasts, and is expected to narrow under non-linear pricing models. This prediction has since been tested directly; see the *Gradient Boosting Extension* (June 2026).
- The Burt-augmented C-series — which retains both categorical dummies and Burt coordinates in the Tweedie design — produces results identical to the Burt-replacing B-series to four decimal places of Gini. The original categorical factors carry no discriminating signal beyond what the Burt coordinates already encode. This informational completeness result justifies restricting the canonical comparison to A-series versus B-series.
- The FFT convolution framework used here imposes practical moment constraints on admissible severity distributions. A candidate screening over twelve distributions confirmed that Burr XII and log-logistic fit the freMTPL2 severity substantially better than lognormal by AIC (improvement of approximately 4,800 points). However, the MLE-fitted parameters place both in a regime of near-infinite skewness ($c \times d = 2.33$ for Burr XII, below the threshold of 3 required for finite third moment), which is incompatible with FFT convolution on a finite grid. Lognormal with empirical per-segment CVs is therefore retained — not because it fits best, but because it has finite moments of all orders. This points to an open question: whether tail-truncated or spliced severity models could reconcile the better distributional fit with the moment requirements of the FFT framework.
- Segment-level severity coefficients of variation, estimated empirically via weighted lognormal MLE restricted to claimants, range from approximately 1.1 to 1.8 across segments — roughly double the conventional fixed assumption of 0.8. The capital impact is material: TVaR 99% increases by approximately €500,000 relative to the fixed-CV assumption. Segment-level severity heterogeneity is both estimable from data and consequential for aggregate tail measurement.
- The coverage-layer analysis reveals a reinsurance-dependent reversal. On a ground-up basis, B5 produces a TVaR 99% approximately €85,000 lower than A5. Under a deductible of 150 EUR this advantage widens to

approximately €550,000. Under a per-claim limit of 12,000 EUR the ranking reverses, with A5 producing a TVaR 99% approximately €478,000 lower. The optimal representation depends not only on the pricing objective but on the reinsurance structure in place.

The most interesting aspect of the work may ultimately not be the individual modelling components themselves, but the interaction between representation geometry, computational structure, and aggregate actuarial behaviour.

Future directions include testing Burt representations in gradient boosting frameworks, extending the approach to high-cardinality categorical variables, and using aggregate capital measures such as VaR or TVaR as additional diagnostics when comparing segmentation designs.

I would be very interested in perspectives on whether similar interactions between representation geometry and collective risk structure have been explored elsewhere in actuarial modelling.

Appendix: Computational Notebook Architecture

Warning: Before running this notebook

This notebook may attempt to install Python packages automatically if they are not already present in your environment. It also creates two cache directories in the working folder: *gmm_sweep_cache/* (BIC sweep results) and *experiment_cache/* (individual pricing experiment results). Both are reused on subsequent runs. Review Cell 1.1 before executing — it lists all required packages and checks whether they are installed. The full dependency list is: *glum, aggregate, scikit-learn, numpy, pandas, matplotlib, scipy*.

The cache directories are safe to delete at any time. The notebook will recompute and recreate them on the next run, at the cost of several hours of computation for the full benchmark.

Notebook Structure

The notebook contains 47 cells organised into ten sections.

Section 1 — Data and Representation. Environment setup, data loading and cleaning, construction of the Burt matrix and policy-level latent coordinates.

Section 2 — Frequency Modeling. Poisson GLM with exposure weighting, calibration and dispersion diagnostics.

Section 3 — Severity Modeling. Severity data merge, distribution candidate screening, and lognormal parameter estimation.

Section 4 — Reusable Modeling Framework. The core functions: build_tweedie_design(), build_cluster_features(), apply_segmentation(), run_pricing_experiment(). These are defined once and reused throughout the benchmark.

Section 5 — Benchmark. K-means elbow diagnostic (Cell 5.1), three-way BIC sweep with disk cache (Cell 5.2), and the full 21-experiment pricing benchmark with per-experiment disk cache (Cell 5.3). This section is the computationally intensive part.

Section 6 — Model Selection and Results. BIC confirmation, three-tier benchmark summary (executive / series comparison / full registry), four visualisation charts including the discrimination–calibration scatter plot.

Section 7 — Aggregate Validation. Helper functions for empirical CV estimation (Cell 7.1), analytic compound Poisson aggregate for the best Gini experiment A5 (Cell 7.2) and the best-calibrated experiment B5 (Cell 7.3). Empirical per-segment CVs are estimated via weighted lognormal MLE from observed claim amounts.

Section 8 — Scenario Layer and Monte Carlo. Two-panel coverage layer analysis comparing A5 and B5 under ground-up, deductible, and per-claim-limit structures (Cell 8.1). Monte Carlo validation using the same empirical CV parameters (Cell 8.2).

Section 9 — QA and Regression Baseline. Regression baseline checks that detect unintended changes to key metrics across runs. Fundamental changes will trigger a python error if UPDATE_BASELINE = False (change when needed).

Section 10 — Reporting Pack. Consolidated executive, diagnostic, and audit views of all benchmark results.

Section 11 — Gradient Boosting Extension. A non-linear pricing layer (LightGBM, Tweedie objective) that replaces the Tweedie GLM while holding the dataset, Burt geometry, and GMM segmentation fixed, with out-of-fold premiums for honest discrimination measurement. Comprises the calibrated GBM fit, an experiment-runner wrapper, the canonical A5/B5/C comparison, and the aggregate and coverage-layer tests under boosting. Added after the original ten-section study; its findings are reported separately in the *Gradient Boosting Extension* (June 2026).

Key Parameters

The following constants at the top of their respective cells control the main modelling choices. All other parameters are derived from these.

GMM_COVARIANCE_TYPE (Cell 5.2) — covariance structure for GMM components. Set to *"full"* throughout. A sensitivity test with *"diag"* confirmed that full covariance with PCA stabilisation is the correct choice for the Burt geometry.

BURT_PCA_VARIANCE_THRESHOLD_PRICING (Cell 4.1) — variance threshold for PCA stabilisation of Burt features before GMM fitting. Set to 0.95, retaining 44 of 61 components. The same threshold is used in the BIC sweep and in the pricing experiments, ensuring consistency.

CALIB_LOWER, CALIB_UPPER (Cell 7.3) — top-decile ratio band for selecting the best-calibrated experiment. Set to 0.95–1.05. The experiment with the highest Gini within this band is selected automatically.

log2, bs (Cells 7.2, 7.3) — FFT grid parameters for the aggregate package. *log2=20* gives 2^{20} = 1,048,576 grid points; *bs=100* sets a bucket size of 100 EUR, covering up to approximately 105M EUR. Increasing *log2* to 21 doubles memory usage and may cause MemoryError on standard laptops.

Experiment Naming Convention

Each of the 21 experiments follows a systematic naming convention:

Series letter — A (classical one-hot), B (Burt-replacing), C (Burt-augmented).

Experiment number — 0 = plain GLM baseline; 1 = K-means simple; 2 = GMM simple; 3 = GMM enriched k=5; 5 = GMM enriched BIC-selected k; ext = K-means enriched k=5.

selection_rule — *fixed* means k was chosen a priori; *fixed_and_bic_selected* means the BIC sweep independently confirms the same k value.

How to Run

Execution proceeds in three phases. In the first phase (Cells 1–4), data is loaded, cleaned, and the Burt coordinates and frequency/severity models are fitted. This takes a few minutes. In the second phase (Cells 5.1–5.3), the BIC sweep and all 21 pricing experiments are run. On the first run this takes approximately two to three hours; on subsequent runs, both the sweep results and the experiment results are loaded from disk cache and complete in seconds. In the third phase (Cells 6–10), results are summarised, visualised, and validated. This is fast.

If the kernel is interrupted during Cell 5.3, any experiments that had already completed are preserved in the cache. On restarting, the notebook will skip cached experiments and continue from where it left off.